Hall Ticket Number:

MR-21

Code: B415CB5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Software Engineering

(Computer Science and Business System)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is meant by software engineering?	2	1	1	1
1. b)	Write the nature of software.	2	1	1	2
1. c)	What is the significance of feasibility studies in requirement engineering?	2	2	2	1
1. d)	Define data model.	2	1	2	1
1. e)	How do we assess the quality of a software design?	2	1	3	2
1. f)	What is meant by software architecture?	2	2	3	1
1. g)	What is the primary goal of a testing strategy?	2	1	4	1
1. h)	Which metric evaluates design complexity?	2	1	4	1
1. i)	Write the ISO 9000 quality standards.	2	1	5	1
1. j)	Define software reliability.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	State and explain various software myths and reality.	5	2	1	1
2. b)	Discuss the capability maturity model integration.	5	3	1	2
	OR				
2. c)	Explain the difference between a software process and a software process model. Suggest two ways in which a software process model might be helpful in identifying possible process improvements. Explain any two process models.	5	4	1	2
2. d)	Describe the evolving role of software.	5	3	1	2
3. a)	Explain various categories of non-functional requirements and their importance in SRS.	5	2	2	2
3. b)	Discuss the Context models and behavioral models.	5	2	2	2
	OR				
3. c)	Identify and explain four types of requirements that may be defined for s computer based system.	5	3	2	1,2
3. d)	Explain the requirements elicitation and analysis process.	5	2	2	2
4. a)	Briefly describe each of the four elements of the design model.	5	2	3	1
4. b)	Explain the sequence diagrams and collaboration diagrams.	5	2	3	1
	OR				
4. c)	Discuss the Design process and design quality.	5	2	3	2
4. d)	Illustrate the architectural styles and patterns.	5	2	3	3
5. a)	Compare and contrast the block box testing and white box testing.	5	2	4	2
5. b)	Discuss the Metrics for Analysis Model and Metrics for Design Model.	5	4	4	1,2
	OR				
5. c)	What is black box testing? What is boundary value Analysis? Explain the technique specifying rules and its usage with the help of an example.	5	2	4	1

P.T.O.

5. d)	Explain the framework for product metrics.	5	3	4	1
6. a)	Compare and contrast the reactive Vs proactive risk strategies.	5	2	5	2
6. b)	Describe the RMMM, RMMM plan.	5	3	5	1,2
	OR				
6. c)	We understand that quality is a "bottom line" issue in all companies. Explain how does software quality assurance fit into an effective project management approach. Describe the activities of software quality assurance team.	5	5	5	4
6. d)	Explain the software risks, risk identification.	5	2	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome